

ARM Physical IP, Inc.

TSMC CL013GFSG (0.13um Generic Process with FSG dielectric)
1.2V SAGE-X(tm) v2.0 Release Notes

2005q3v1 (09/16/2005)

Impact on Designs Using 2004q1v1

This section describes the impact of this release on current designs which are using the previous release.

*** This is a mandatory update. ***

ARM Update Recommendation:

Install version 2005q3v1. Discontinue using 2004q1v1.

- o GDS2 has not changed
- o P&R footprints have not changed.
- o Updated the three_state equation in the .lib for RF1R1WX1 cell.
- o Updated the Enable and Latch statements in the .tlf for RF1R1WX1 and RF2R1WX1 cells.
- o Updated power for the following cells to add fall_power model to passive arc CK (when !D && E && !RN).
- o Timing and power have not changed for all other cells.
- o Updated LEF and Apollo tf to design rule document T-013-L0-DR-001 Rev 2.1.
- o Updated sdf post processor for PrimeTime.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o Design Rule Document
 - T-013-L0-DR-001 Rev 2.1 03-31-04
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S_8M.21c (VER 2.1c, 6/14/2004)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V1.3c, 2002/06/25
- o Layer Map File
 - TSMC T-013-L0-LE-001 v2.1a 5/24/2004
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-L0-SP-004 V1.3
- o Process Models
 - 1.3P1 Jan 11, 2002

EDA Support

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to ARM EDA Package 3.0.1.

* Cadence NC-Sim (Verilog)

- 3.1
- o Cadence Verilog-XL (Verilog)
 - 3.10-p001
- * Synopsys VCS (Verilog)
 - 5.1
- o Mentor ModelSim (Verilog)
 - 5.5
- o Cadence NC-VHDL (VHDL)
 - 3.1
- o Synopsys VSS (VHDL)
 - 2000.12
- * Mentor ModelSim (VHDL)
 - 5.5
- * Simulation Model SDF Compatibility
 - SDF 2.1
- * Synopsys Design Compiler (Synthesis, Delay Calculation)
 - 2000.11
- o Synopsys Module Compiler
 - 2000.11
- * Cadence BuildGates/PKS
 - 4.0-s009
- o Synopsys Physical Compiler
 - 2000.11
- o Synopsys Chip Architect
 - 2000.11
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2001.06
- * Cadence Pearl (Static Timing, Delay Calculation)
 - 4.3-s086 (TLF 4.1)
- * Cadence NanoRoute (SoC Encounter)
 - 2.1 (3.1) with LEF 5.4 & .lib
- o Cadence Silicon Ensemble
 - 5.4-USR4 with LEF 5.4 & TLF 4.1
- * Synopsys Apollo-II (FRAM, CEL verified; TIM, PWR, CLF supported)
 - 2000.2
- * Mentor Graphics Calibre (GDSII, CDL)
 - d.1
- * Cadence Composer (Schematic Entry)
 - DFII 4.4.3.72
- o Synopsys DFT Compiler
 - 2000.11
- * Synopsys Power Compiler
 - 2000.11
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- o Synopsys HSPICE
 - 2003.03

Technology Implementation

 This section provides information on items that may not be included in the foundry documentation.

Characterization

- Timing and power views characterized at the following conditions:
 fast P/V/T = FF/1.32V/-40C

fastz P/V/T = FF/1.32V/0C
typical P/V/T = TT/1.20V/25C
slow P/V/T = SS/1.08V/125C

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
*** This is a mandatory tapeout layer ****

TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

Methodology

This section describes design flow issues that may be applicable to this release.

o DRC

- The FILL1 cell will fail a variety of DRC errors when run on FILL1 individually. These errors will disappear at chip-level DRC due to interaction with neighboring cells:

NP.A.1, Min. area of NP 0.25
PP.A.1, Min. area of PP 0.25
OD.A.1, Min. area of stand-alone OD region 0.122

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o There are no known problems with this release.

2004q1v1 (03/01/2004)

Impact on Designs Using 2003q4v1

This section describes the impact of this release on current designs which are using the previous release.

*** This is an optional update. ***

Artisan Update Recommendation:

Install version 2004q1v1. Re-evaluate timing simulations before tape-out.
Re-run Verilog simulation if SMDFFHQ* cells are used.
Discontinue using 2003q4v1.

- o LEF files have been updated to version 5.4 and technology section has been optimized to support NanoRoute and the Encounter platform.
- o GDS2 has not changed.
- o P&R footprints have not changed.
- o All timing corners updated CMPR42* sdf conditions statements to be consistent with the Verilog models.

- o Timing has been updated. Timing numbers were recharacterized with a newer version of hspice simulator in order to support additional model views. As a result, very minor timing variations were introduced into the timing models as compared to the previous release.
- o Updated layout area for ACCSHCINX2 and ACCSHCINX4 in tlf.
- o Verilog models updated to make timescale consistent throughout library, and to update SMDFFHQ* path delays.
- o Apollo *.tf and libraries have updated METAL1 current density value.
- o Databook updated to add cells ACHCIN and ACHCON. Also updated TLATNTSCA and updated equations to BENC.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.5 08-01-02
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S_8M.15f (VER 1.5f, 4/14/2003)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V1.3c, 2002/06/25
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004 V1.3
- o Process Models
 - 1.3P1 Jan 11, 2002

EDA Views

This section lists the EDA tools and versions used to verify this product release. This set of tools and versions corresponds to Artisan EDA Package 3.0.1.

- o Cadence NC-Verilog
 - 3.1
- o Synopsys VCS
 - 5.1
- o MTI/Mentor ModelSim (VHDL)
 - 5.5
- o Cadence BuildGates with Low Power Synthesis
 - v4.0-s009
- o Synopsys Design Compiler with Power Compiler
 - 2000.11
- o Synopsys Physical Compiler
 - 2000.11
- o Synopsys PrimeTime
 - 2001.06
- o Cadence Pearl (TLF 4.1)
 - 4.3-s086
- o Cadence NanoRoute (SoC Encounter)
 - 2.1 (3.1)
- o Synopsys Apollo II
 - 2000.2
- o Adobe Acrobat Reader
 - 4.0

- o Mentor Calibre
 - d.1
- o Synopsys HSPICE
 - 2003.03

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

Characterization

- Timing and power views characterized at the following conditions:
 - fast P/V/T = FF/1.32V/-40C
 - fastz P/V/T = FF/1.32V/0C
 - typical P/V/T = TT/1.20V/25C
 - slow P/V/T = SS/1.08V/125C

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
 *** This is a mandatory tapeout layer ****

TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

Methodology

This section describes design flow issues that may be applicable to this release.

o DRC

- The FILL1 cell will fail a variety of DRC errors when run on FILL1 individually. These errors will disappear at chip-level DRC due to interaction with neighboring cells:
 - NP.A.1, Min. area of NP 0.25
 - PP.A.1, Min. area of PP 0.25
 - OD.A.1, Min. area of stand-alone OD region 0.122

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o There are no known problems with this release.

 2003q4v1 (10/15/2003)

Impact on Designs Using 2002q1v3

This section describes the impact of this release on current designs which are using the previous release.

*** This is a mandatory update. ***

Artisan Update Recommendation:

Install version 2003q4v1. Discontinue using 2002q1v3.

- o The following cells have changed P&R footprint and gds2:
AND3X6 AND3X8 AND4X8 BUFX16 BUFX20 CLKBUFX20
INVX12 TBUFX6
If using any of the above cells, replace the cell and check placement and routing. These cells have been modified to improve reliability due to electromigration. Timing values for these cells has not changed.
- o Overall library timing values have not changed except cell INVX1 in Cadence typical.tlf.
- o Overall library power values have not changed.

Other Library Enhancements:

- o Antenna ratio has been updated in tsmc13*lm_antenna.lef and antenna*.clf to comply with DRC VER 1.5f.
- o Changed metal overlap of via value to be 0.05um in tsmc13*lm.lef, tsmc13*lm.plib, tsmc13*lm.pdb to be consistent with Apollo tf.
- o Added more LEQ relationships for Apollo leq.clf file.
- o Apollo tech files tsmc13*lm.tf have been updated:
 - Updated capacitance models in tsmc13*lm.tf.
 - Added OD2 layer.
 - Added max current density value.
- o Apollo files gdsInLayer.map and gdsOutLayer.map have been changed as follows:
 - Added OD2 layer.
 - Changed "NDIFF" and "PDIFF" layers to "DIFF" layer.
- o Synopsys files *.lib and *.db have been updated as follows:
 - The current_unit has changed from "uA" to "mA".
- o The Verilog models tsmc13.v and tsmc13_neg.v have been updated as follows:
 - udp_sedfft and udp_edfft changed. Affects cells SDDFFTRX*, SEDFFTRX*, DFFTRX*, EDFFTRX*
 - UDP changes to udp_bmx to remove pessimism. Cells affected are: BMXX2, BMXX4, BMXIX2, BMXIX4
 - Changed timing place holder for FF,latch to eliminate false warning during unit simulation.
 - Changed timescale from "1ns/10ps" to "1ns/1ps".

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o Design Rule Document
 - T-013-L0-DR-001 Rev 1.5 08-01-02
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S_8M.15f (VER 1.5f, 4/14/2003)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V1.3c, 2002/06/25
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document

- T-013-LO-SP-004 V1.3
- o Process Models
 - 1.3P1 Jan 11, 2002

EDA Views

This section lists the EDA tools and versions used to verify this product release. This set of tools and versions corresponds to Artisan EDA Package 3.0.

- o Cadence NC-Verilog
 - 3.1
- o Synopsys VCS
 - 5.1
- o MTI/Mentor ModelSim (VHDL)
 - 5.5
- o Cadence BuildGates with Low Power Synthesis
 - v4.0-s009
- o Synopsys DesignCompiler with DesignPower
 - 2000.11
- o Synopsys PhysicalCompiler
 - 2000.11
- o Synopsys PrimeTime
 - 2001.06
- o Cadence Pearl (TLF 4.1)
 - 4.3-s086
- o Cadence Silicon Ensemble
 - 5.3
- o Synopsys ApolloII
 - 2000.2
- o Adobe Acrobat Reader
 - 4.0
- o Mentor Graphics Calibre
 - d.1
- o Synopsys Hspice
 - H98.2

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

Characterization

- Timing and power views characterized at the following conditions:

fast	P/V/T = FF/1.32V/-40C
fastz	P/V/T = FF/1.32V/0C
typical	P/V/T = TT/1.20V/25C
slow	P/V/T = SS/1.08V/125C

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
 *** This is a mandatory tapeout layer ****

TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:

<http://online.tsmc.com.tw/>

Methodology

This section describes design flow issues that may be applicable to this release.

o DRC

- The FILL1 cell will fail a variety of DRC errors when run on FILL1 individually. These errors will disappear at chip-level DRC due to interaction with neighboring cells:

NP.A.1, Min. area of NP 0.25

PP.A.1, Min. area of PP 0.25

OD.A.1, Min. area of stand-alone OD region 0.122

- o Attention: For Cadence customers using the Encounter platform, including First Encounter and/or NanoRoute Ultra, please contact your Cadence applications engineer to obtain Encounter tuned LEF files for optimal results.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o There are no known problems with this release.

2002q1v3 (11/19/2002)

Changes

- o Layouts have not changed.
- o P&R Footprints have not changed.
- o Timing & Power Data have not changed.

Front-end

- o Verilog
 - UDP changes to udp_sedfft and udp_edfft. Cells affected are: SDDFTRX*, SEDFFTRX*, DFFTRX*, EDFDTRX*
- o Apollo tech file
 - Changed power unit from "mw" to "pw" for consistency with .lib files.
 - Changed precision on current, power, capacitance.
 - Added maxNumRowsNonTurning to fat via definitions
- o Apollo database
 - Rebuild TIM and PWR database with new units and precision

Back-end

- o No changes to back-end views in this release

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.5 08-01-02
- o DRC Command File

- 0.13um CALIBRE DRC COMMAND FILE - CL13S_4M.15a.1 (VER 1.5a.1,
07/29/2002)

- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V1.3c, 2002/06/25
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004
- o Process Models
 - 1.3P1 Jan 11, 2002
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o ModelSim
 - 5.5
- o BuildGates with Low Power Synthesis
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PhysicalCompiler
 - 2000.11
- o PrimeTime
 - 2000.06
- o Pearl (TLF 4.1)
 - 4.3-s086
- o Silicon Ensemble
 - 5.3
- o ApolloII
 - 2000.2
- o Acrobat Reader
 - 4.0
- o Calibre
 - d.1
- o Hspice
 - H98.2

Comments

- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2002q1v2 (07/15/2002)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.4 06-20-02
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S_8M.14a.2 (VER 1.4a.2,
06/03/2002)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004

- o Process Models
 - 1.3P1 Jan 11, 2002
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o ModelSim
 - 5.5
- o BuildGates with Low Power Synthesis
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PhysicalCompiler
 - 2000.11
- o PrimeTime
 - 2000.06
- o Pearl (TLF 4.1)
 - 4.3-s086
- o Silicon Ensemble
 - 5.3
- o ApolloII
 - 2000.2
- o Acrobat Reader
 - 4.0
- o Calibre
 - d.1
- o Hspice
 - H98.2

Front-end

- o LEF
 - Updated RC values to correspond with HSPICE FSG interconnect model version 1.3
- o TLF
 - Updated function statement in TLATNTSCA and TLATNCA cells
- o Databook
 - Corrected typos in footer
 - Added voltage and temperature above delay tables
- o Synopsys .LIB
 - Added max_transition to CLK pin for all flip-flops and latches
 - Updated header variables:
 - slew_lower_threshold_pct_fall : 30.0;
 - slew_upper_threshold_pct_fall : 70.0;
 - slew_lower_threshold_pct_rise : 30.0;
 - slew_upper_threshold_pct_rise : 70.0;
 - slew_derate_from_library : 0.5;
- o Apollo tech file
 - Changed METAL1 minSpacing from 0.22 to 0.18
 - Removed stubSpacing and stubThreshold lines
 - Changed fatTblSpacing for 6lm (0.22 -> 0.18)
 - Updated RC values to correspond with HSPICE FSG interconnect model version 1.3

Back-end

- o No changes to back-end views in this release
- o TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2002q1v1 (04/25/2002)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.2 11-23-01
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.12a (VER 1.2a, 11/12/2001)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004
- o Process Models
 - 1.3P1 Jan 11, 2002
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o ModelSim
 - 5.5
- o BuildGates with Low Power Synthesis
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PhysicalCompiler
 - 2000.11
- o PrimeTime
 - 2000.06
- o Pearl (TLF 4.1)
 - 4.3-s086
- o Silicon Ensemble
 - 5.3
- o ApolloII
 - 2000.2
- o Acrobat Reader
 - 4.0
- o Calibre
 - d.1
- o Hspice
 - H98.2

Front-end

- o EDFFTRX4 & SEDFFTRX4
 - Rebuilt timing models for typical EDFFTRX4 and slow SEDFFTRX4
- o Apollo tech file
 - Changed DesignRule for via-to-viaBlockage spacing from 0.22 to 0.29.
 - Defined fat vias with more than 1 cut for VIAx.R2 rule.
 - Added minEnclosedArea rule to metal layers.
 - Updated resistance values
- o LEF
 - Added MINFEATURE property

- Changed metal overlap of vias from 0.145 to 0.155 on long side to avoid short net minimum area violations for neighboring vias

Back-end

- o No changes to back-end views in this release
- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2002q1v0 (03/08/2002)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.2 11-23-01
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.12a.preqa3 (VER 1.2a, 11/12/2001)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004
- o Process Models
 - 1.3P1 Jan 11, 2002
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o ModelSim
 - 5.5
- o BuildGates with Low Power Synthesis
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PhysicalCompiler
 - 2000.11
- o PrimeTime
 - 2000.11
- o Pearl (TLF 4.1)
 - 4.3-s086
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.7
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o VHDL
 - Changed polarity of enable signals in RF1R1W cell.
 - Consolidated lines in udp_tlatrf to improve readability.

- Added a udp_tlatrf2. Not used in this model file.
- Reduced pessimism in udp_edfft and udp_sedfft where E=0 and RN or SN is X.
- Added recovery time for cells with asynchronous set and reset signals
- o Databook
 - Added DFFQ to databook
- o Synopsys
 - Set "dont_use" and "dont_touch" properties on RFRD cells. These cells should be instantiated by hand.
 - Added PhysicalCompiler models to release (.plib)
- o ACHCINX4 area
 - Added layout area to fast corner models (Synopsys, TLF, ALF)
- o Characterization (Synopsys, TLF, ALF, CLF, databook)
 - Recharacterized library using spice models version 1.3P1
 - Added fourth corner (fastz) to characterization

fast	P/V/T = FF/1.32V/-40C
fastz	P/V/T = FF/1.32V/0C
typical	P/V/T = TT/1.20V/25C
slow	P/V/T = SS/1.08V/125C

Back-end

- o CDL (lvs netlist)
 - Modified area parameter in ANTENNA cell to support Hercules LVS deck
- o Layouts (gds2 and Apollo CEL)
 - Modified NWELL and Implant layers in all layouts to make layouts compatible with TSMC's HiVt process. These changes have no impact on performance or place & route results.
 - Modified field POLY layer in AFCSHCINX4 layout to make layout compatible with TSMC's HiVt process. This change has no impact on performance or place & route results.
 - Modified METAL1 geometries in BUFX20, TLATNCAX16 and TLATNCAX20 for design rule VIA1.R.2 compatibility in TSMC's version 1.2 design rules. This change may result in a change of place & route results, but no change to timing characteristics.
- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2001q4v1 (11/13/2001)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.1
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.11a (VER 1.1a, 06/28/2001)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004 V1.0
- o Process Models
 - 1.0 Oct. 31, 2000
- o NC-Verilog

- 3.1
- o VCS
 - 6.0
- o ModelSim
 - 5.5
- o BuildGates with Low Power Synthesis
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PrimeTime
 - 2000.11
- o Pearl
 - 4.3-s086
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.7
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o Fast corner timing models (.lib, .tlf, .alf)
 - Added area value for ACHCINX4 cell
- o LEF
 - Added "VERSION 5.4 ;" to antenna LEF
 - Added "CLASS CORE ANTENNACELL ;" to ANTENNA cell in antenna LEF
 - Added "CLASS CORE SPACER ;" to FILL cells in antenna LEF
 - Increased pitch on metal layer N-1

Back-end

- o No changes to back-end in this release
- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2001q4v0 (10/14/2001)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.1
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.11a (VER 1.1a, 06/28/2001)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004 V1.0
- o Process Models
 - 1.0 Oct. 31, 2000

- o NC-Verilog
 - 3.1
- o VCS
 - 6.0
- o ModelSim
 - 5.5
- o BuildGates
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PrimeTime
 - 2000.11
- o Pearl
 - 4.3-s086
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.7
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o Redesigned and recharacterized the following cells:
 - AFCSHCINX2 AFCSHCINX2 AFHCINX2 AFHCINX2 AHHCINX2
 - AFCSHCINX4 AFCSHCINX4 AFHCINX4 AFHCINX4 AHHCINX4
- o StarDC
 - Dropped support for StarDC model views.
- o ALF
 - Added support for Ambit ALF model view. NOTE: ALF models are currently under Beta-test. These models are subject-to-change.
- o LEF
 - Modified LEF to support via stacking

Back-end

- o Redesigned and recharacterized the following cells:
 - AFCSHCINX2 AFCSHCINX2 AFHCINX2 AFHCINX2 AHHCINX2
 - AFCSHCINX4 AFCSHCINX4 AFHCINX4 AFHCINX4 AHHCINX4
- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

 2001q3v1 (09/18/2001)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.1
- o DRC Command File
 - 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.11a (VER 1.1a, 06/28/2001)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14

- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-LO-SP-004 V1.0
- o Process Models
 - 1.0 Oct. 31, 2000
- o NC-Verilog
 - 3.1
- o VCS
 - 6.0
- o ModelSim
 - 5.5
- o BuildGates
 - v4.0-s009
- o DesignCompiler with DesignPower
 - 2000.11
- o PrimeTime
 - 2000.11
- o Pearl
 - 4.3-s086
- o Star-DC
 - 1998.2
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.1
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o The following 10 SAGE-X cells have the 'don't use' property enabled in the .lib files. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release.

AFCSHCINX2 AFCSHCONX2 AFHCINX2 AFHCONX2 AHHCINX2
 AFCSHCINX4 AFCSHCONX4 AFHCINX4 AFHCONX4 AHHCINX4

Back-end

- o The following 10 SAGE-X cells have been removed from the gds2 and apollo views. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release.

AFCSHCINX2 AFCSHCONX2 AFHCINX2 AFHCONX2 AHHCINX2
 AFCSHCINX4 AFCSHCONX4 AFHCINX4 AFHCONX4 AHHCINX4

2001q3v0 (08/25/2001)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev 1.1
- o DRC Command File

- 0.13um CALIBRE DRC COMMAND FILE - CL13S1.0V+1.2V+2.5V_8M.11a (VER 1.1a, 06/28/2001)
 - o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
 - o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
 - o Process Document
 - T-013-LO-SP-004 V1.0
 - o Process Models
 - 1.0 Oct. 31, 2000
 - o NC-Verilog
 - 3.1
 - o VCS
 - 6.0
 - o ModelSim
 - 5.5
 - o BuildGates
 - v4.0-s009
 - o DesignCompiler with DesignPower
 - 2000.11
 - o PrimeTime
 - 2000.11
 - o Pearl
 - 4.3-s086
 - o Star-DC
 - 1998.2
 - o Silicon Ensemble
 - 5.3.60
 - o ApolloII
 - 2000.2.3.4.0.1
 - o Acrobat Reader
 - 4.0
 - o Calibre
 - v8.7_30.3
 - o Hspice
 - H98.2

Front-end

- o Redesigned and recharacterized the following cells:

ACCSHCINX2 ACCSHCONX4 ACHCONX2 AFCSHCINX4 AFHCINX2 AFHCONX4
 ACCSHCINX4 ACHCINX2 ACHCONX4 AFCSHCONX2 AFHCINX4
 ACCSHCONX2 ACHCINX4 AFCSHCINX2 AFCSHCONX4 AFHCONX2
- o Sunrise
 - Discontinued support for Sunrise
- o Synopsys
 - Recharacterized D setup time in slow SEDFFHQX1 to fix bad data point
- o TLF
 - Recharacterized D setup time in slow SEDFFHQX1 to fix bad data point
- o ALF
 - Added ALF view to release for Ambit support
- o StarDC
 - Recharacterized D setup time in slow SEDFFHQX1 to fix bad data point
- o Apollo
 - Changed "PDIFF" layer name to "DIFF" in tech files
 - Changed "flipflop" attribute to "clock" for TLAT*C* cells

o Known Problems

- Apollo - Current versions of Apollo may have trouble routing a DRC/LVS clean design for TSMC's 0.13um design rules. Artisan and TSMC are working with Avant! to develop a clean place & route flow for Apollo. Apollo may also report several false metal1 spacing errors of this form:

```
Met1 Spacing          100      metal1 minimum spacing = 0.22
Limit of the maximum errors has been reached.
Further checking is skipped!
```

- SiliconEnsemble - SE may report some false metal1 spacing violations during SRoute. Likewise, it may create some occasional metal1 spacing violatons. Artisan is working with Cadence to develop a clean place & route flow for SiliconEnsemble.
- SAGE-X Cells - The following SAGE-X cells have the "dont_use" property enabled in the .LIB file for DesignCompiler. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be recharacterized in an upcoming release:

```
AFCSHCINX2 AFCSHCONX2 AFHCINX2   AFHCONX2   AHHCINX2
AFCSHCINX4 AFCSHCONX4 AFHCINX4   AFHCONX4   AHHCINX4
```

Back-end

- o Redesigned and recharacterized the following cells:

```
ACCSHCINX2 ACCSHCONX4 ACHCONX2   AFCSHCINX4 AFHCINX2   AFHCONX4
ACCSHCINX4 ACHCINX2   ACHCONX4   AFCSHCONX2 AFHCINX4
ACCSHCONX2 ACHCINX4   AFCSHCINX2 AFCSHCONX4 AFHCONX2
```

- o The following cell layouts were changed for cosmetic reasons, and will not affect yield, timing, or place & route results:

```
AND2X1      DFFNSRXL   DFFSRHQX1 DFFSRHQX8 SEDFFHQX1 SEDFFHQX2
DFFNSRX2    DFFQXL     DFFSRHQX4 NOR3X1    SDFFSX1
```

- o The following cell layouts were changed to reflect changes made in Calibre DRC version 1.1a for the 0.13um process:

```
TLATNCAX8 -- Design rule PP.R.1
SDFFHQX1  -- Design rule LDP.0.2A
```

These cells were modified without changing metal1 or transistor sizes, timing and place & route results will not be impacted.

- o TSMC Calibre DRC/LVS/LPE, SPICE models
 - Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

2001q1v1 (03/23/2001)

QA

- o Design Rule Document
 - T-013-LO-DR-001 Rev1.0

- o DRC Command File
 - 0.13um Calibre DRC COMMAND FILE - CL13S1.0V+1.2V+3.3V_6M.10b (VER 1.0b,12-21-00)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-L0-SP-004 V1.0
- o Process Models
 - 1.0 Oct. 31, 2000
 - The product is characterized using TSMC's spice model v1.0. The timing is expected to be more conservative than the v1.1 spice model. Users should exercise caution in hold time margins when using this product. Artisan will release a new version of this product within the next six months based on spice model v1.1. Please contact your TSMC account manager for more detail on the spice models.
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o VSim
 - 5.4a
- o DesignCompiler with DesignPower
 - 1999.10
- o PrimeTime
 - 1999.10
- o Pearl
 - 4.3-p025
- o Star-DC
 - 1998.2
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.1
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o Apollo
 - Added Resistance and Capacitance values to apollo tech files.
- o LEF
 - Added Resistance and Capacitance values to LEF
- o Known Problems
 - Apollo - Current versions of Apollo may have trouble routing a DRC/LVS clean design for TSMC's 0.13um design rules. Artisan and TSMC are working with Avant! to develop a clean place & route flow for Apollo. Apollo may also report several false metal1 spacing errors of this form:

Met1 Spacing 100 metal1 minimum spacing = 0.22
 Limit of the maximum errors has been reached.
 Further checking is skipped!

- SiliconEnsemble - SE may report some false metal1 spacing violations during SRoute. Likewise, it may create some occasional metal1 spacing violatons. Artisan is working with Cadence to develop a clean place & route flow for SiliconEnsemble.
- SAGE-X Cells - The following SAGE-X cells have the "dont_use" property enabled in the .LIB file for DesignCompiler. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release:

ACCSHCINX2 ACCSHCONX4 ACHCONX2 AFCSHCINX4 AFHCINX2 AFHCONX4
 ACCSHCINX4 ACHCINX2 ACHCONX4 AFCSHCONX2 AFHCINX4
 ACCSHCONX2 ACHCINX4 AFCSHCINX2 AFCSHCONX4 AFHCONX2

Back-end

o GDS2

- Removed GDS2 for the following 8 SAGE-X cells. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release:

ACCSHCINX2 ACCSHCINX4 ACCSHCONX2 ACCSHCONX4
 AFCSHCINX2 AFCSHCINX4 AFCSHCONX2 AFCSHCONX4

o Apollo

- Removed CEL views for the following 8 SAGE-X cells. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release:

ACCSHCINX2 ACCSHCINX4 ACCSHCONX2 ACCSHCONX4
 AFCSHCINX2 AFCSHCINX4 AFCSHCONX2 AFCSHCONX4

o TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:
<http://online.tsmc.com.tw/>

 2001q1v0 (03/15/2001)

QA

- o Design Rule Document
 - T-013-L0-DR-001 Rev1.0
- o DRC Command File
 - 0.13um Calibre DRC COMMAND FILE - CL13S1.0V+1.2V+3.3V_6M.10b (VER 1.0b,12-21-00)
- o LVS Command File
 - TSMC 0.13UM CALIBRE ERC/LVS COMMAND FILE V0.3a, 2000/12/14
- o Process Name
 - 0.13um LOGIC SALICIDE FSG IMD (1P8M,1.2V/2.5V) (CL013G)
- o Process Document
 - T-013-L0-SP-004 V1.0
- o Process Models
 - 1.0 Oct. 31, 2000
- o NC-Verilog
 - 3.1
- o VCS
 - 5.1
- o VSim
 - 5.4a

- o DesignCompiler with DesignPower
 - 1999.10
- o PrimeTime
 - 1999.10
- o Pearl
 - 4.3-p025
- o Star-DC
 - 1998.2
- o Silicon Ensemble
 - 5.3.60
- o ApolloII
 - 2000.2.3.4.0.1
- o Acrobat Reader
 - 4.0
- o Calibre
 - v8.7_30.3
- o Hspice
 - H98.2

Front-end

- o Initial release
- o Characterization (Synopsys, TLF, StarDC, CLF, Databook)
 - Timing and power views characterized at the following conditions:
 - fast P/V/T = FF/1.32V/-40C
 - typical P/V/T = TT/1.20V/25C
 - slow P/V/T = SS/1.08V/125C
- o Known Problems
 - Apollo - Current versions of Apollo may have trouble routing a DRC/LVS clean design for TSMC's 0.13um design rules. Artisan and TSMC are working with Avant! to develop a clean place & route flow for Apollo. Apollo may also report several false metal1 spacing errors of this form:

Met1 Spacing 100 metal1 minimum spacing = 0.22

Limit of the maximum errors has been reached.

Further checking is skipped!
 - SiliconEnsemble - SE may report some false metal1 spacing violations during SRoute. Likewise, it may create some occasional metal1 spacing violatons. Artisan is working with Cadence to develop a clean place & route flow for SiliconEnsemble.
 - RC values - Resistance and Capacitance values, needed for timing driven place & route, are not available for this release; however, they will be added to an upcoming release soon.
 - SAGE-X Cells - The following SAGE-X cells have the "dont_use" property enabled in the .LIB file for DesignCompiler. DO NOT USE THESE CELLS IN YOUR DESIGN. They will be redesigned in an upcoming release:

```

ACCSHCINX2 ACCSHCONX4 ACHCONX2   AFCSHCINX4 AFHCINX2 AFHCONX4
ACCSHCINX4 ACHCINX2   ACHCONX4   AFCSHCONX2 AFHCINX4
ACCSHCONX2 ACHCINX4   AFCSHCINX2 AFCSHCONX4 AFHCONX2

```

Back-end

- o Initial release

- o TSMC Calibre DRC/LVS/LPE, SPICE models

- Calibre rules files and SPICE models available at TSMC's website:

- <http://online.tsmc.com.tw/>
